

ANTANANARIVO IVATO, Madagascar

WMO: 670830

Lat: 18.797S

Lon: 47.479E

Elev: 4198

StdP: 12.60

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	46.4	48.2	40.8	43.8	67.3	43.0	47.8	62.8	18.3	63.2	16.7	63.1	1.8	160	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	19.5	86.1	67.1	84.3	67.4	82.7	67.4	73.8	79.8	72.5	79.0	71.1	78.0	6.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	138.0	77.0	70.2	130.0	75.4	69.5	127.1	74.6	40.6	79.7	39.2	79.0	38.0	78.1	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	15.1	13.6	DB	41.4	89.9	2.0	1.4	40.0	91.0	38.8	91.8	37.7	92.6	36.2	93.6	(4)
(5)			WB	41.0	77.1	2.1	2.1	39.5	78.6	38.3	79.8	37.1	81.0	35.6	82.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.3	71.9	71.8	71.3	69.2	65.6	61.2	59.7	61.1	64.4	68.6	71.0	72.2		
	DBStd	5.31	2.12	2.18	2.25	2.48	2.99	2.97	2.75	2.86	3.40	3.48	2.91	2.18		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	503	0	0	0	2	29	119	166	124	52	9	2	0		
	CDD50	6316	679	610	662	577	482	335	300	344	431	577	629	690		
	CDD65	1345	214	190	197	130	46	4	1	4	33	121	180	225		
	CDH74	7155	921	804	835	574	214	30	15	70	353	973	1158	1208		
	CDH80	1242	143	118	109	53	9	0	3	46	238	281	242			
(14) Wind	WSAvg	5.9	5.8	5.8	5.7	5.4	4.9	5.3	6.0	6.6	7.1	6.9	6.0	5.3		
(15) Precipitation	PrecAvg	55.7	11.9	10.6	7.7	2.0	0.7	0.2	0.3	0.3	0.3	2.4	6.3	12.0		
	PrecMax	72.4	31.3	24.9	20.0	6.2	6.1	1.3	1.5	1.2	2.4	12.2	14.3	28.3		
	PrecMin	33.1	3.1	1.2	1.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.1		
	PrecStd	10.6	6.5	5.4	4.0	1.7	1.1	0.3	0.4	0.3	0.4	2.2	3.1	5.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.9	85.6	84.5	83.9	80.9	77.2	75.6	79.1	84.3	88.2	88.1	87.4		
		MCWB	69.7	70.7	69.8	67.2	65.4	62.0	60.4	59.6	62.3	65.4	66.5	68.9		
	2%	DB	83.1	82.8	82.6	81.4	78.7	74.3	73.1	75.5	80.9	85.4	85.8	84.5		
		MCWB	69.5	69.8	69.8	66.9	64.5	61.0	58.6	59.5	61.1	64.7	66.1	68.4		
	5%	DB	81.1	80.9	80.8	79.2	76.6	71.9	70.3	73.2	78.4	82.5	83.5	82.6		
		MCWB	69.3	69.3	68.9	66.5	63.8	60.7	58.4	58.7	61.0	63.8	66.2	68.5		
	10%	DB	79.1	79.0	78.9	77.4	73.8	69.7	68.1	70.2	75.2	79.5	80.8	80.6		
		MCWB	68.9	68.7	68.4	66.1	63.3	59.9	57.8	58.3	60.4	63.2	65.8	68.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.0	74.9	74.6	72.3	69.9	66.9	64.7	64.7	66.8	70.9	73.3	75.1		
		MCDB	80.1	80.1	80.6	78.5	76.5	72.2	69.8	71.8	76.9	81.4	80.7	81.4		
	2%	WB	73.4	72.9	72.7	70.2	67.5	64.2	62.1	62.1	64.6	68.4	70.9	72.9		
		MCDB	78.8	79.1	79.1	76.7	73.8	69.9	68.4	70.5	74.5	77.7	79.6	79.8		
	5%	WB	71.7	71.2	71.1	68.7	66.0	62.5	60.3	60.8	63.2	66.7	69.2	71.0		
		MCDB	77.6	77.8	77.7	75.6	72.6	68.3	67.4	69.2	73.1	76.4	77.7	78.2		
	10%	WB	70.1	69.7	69.5	67.3	64.3	60.9	58.9	59.4	61.7	65.0	67.6	69.4		
		MCDB	76.3	76.5	75.8	74.4	71.1	67.0	65.7	67.3	71.3	74.2	75.8	76.6		
(35) Mean Daily Temperature Range	MDBR		17.1	16.9	17.2	18.4	18.6	18.7	18.8	20.3	22.7	23.6	21.9	19.5		
	5% DB	MCDBR	19.8	19.5	19.5	20.8	21.9	21.9	22.2	24.5	27.6	27.5	25.1	21.8		
		MCWBR	9.4	9.3	9.4	9.5	10.3	11.0	10.9	10.8	10.9	10.3	9.8	9.5		
	5% WB	MCDBR	17.3	17.7	18.1	19.0	19.3	19.5	19.7	21.1	23.6	24.0	21.7	19.7		
		MCWBR	10.2	10.1	10.2	10.4	10.8	11.6	11.3	11.5	11.4	10.9	10.5	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.333	0.327	0.318	0.299	0.284	0.268	0.272	0.285	0.316	0.363	0.369	0.345		
	taud		2.668	2.702	2.730	2.780	2.794	2.803	2.736	2.630	2.500	2.376	2.416	2.562		
	Ebn at Noon		321	319	315	310	304	304	305	310	310	304	307	317		
	Edh at Noon		31	30	28	25	23	22	23	28	34	41	40	35		
(44) All-Sky Solar Radiation	RadAvg		1848	1837	1769	1658	1406	1280	1306	1568	1913	2074	2052	2027		
	RadStd		177	157	84	85	67	58	69	91	104	106	120	137		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.52	N/A	N/A	+0.86	+2.52	+2.99	N/A	-112	+295	+148
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page